Laboratory 3

Variant 2, Group 32

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1. INTRODUCTION

In this assignment,we use a genetic algorithm (GA) to solve a continuous optimization problem in two dimensions. The goal in this task is to minimize the Himmelblau's function, which is a multimodal benchmark that contains four distinct global minima of equal value.

The decision variables are encoded as binary chromosomes that are decoded into real-valued parameters within a predefined range. Each individual in the population represents a candidate solution, whose fitness is given directly by the value of the objective function. The algorithm iteratively improves the population using tournament selection, one-point crossover, and random mutation.

Himmelblau's function is defined as

$$f(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$

and is commonly used to test whether an optimizer can escape local minima and reliably locate one of several equivalent global optima. Because the landscape is highly non-convex, gradient-based methods are sensitive to initialization, which makes population-based metaheuristics such as genetic algorithms particularly attractive.

The implementation follows the provided template.py skeleton and is designed to be general: the decoding procedure and the objective evaluation work for an arbitrary number of dimensions, although in this assignment only two variables are used. The report focuses on analysing how key hyperparameters of the GA—mutation probability, tournament size, population size, and crossover probability—affect convergence speed, solution quality, and robustness across multiple independent runs.

2. IMPLEMENTATION

In this assignment, the task is to minimize Himmelblau's function using a binary-coded genetic algorithm. The objective function is defined as previously mentioned and real-valued decision variables constrained to the bounded domain [-5, 5].

The optimization goal is to find values of (x_1,x_2) that yield the smallest possible function value, i.e. approximate one of the four global minima of Himmelblau's function within this search space. Candidate solutions are represented as binary chromosomes that are decoded into continuous variables in [−5,5], evaluated by the objective function, and iteratively improved through tournament selection, one-point crossover, and mutation until the algorithm converges to a region of low function value.

2.1 Genetic Algorithm Setup

The genetic algorithm operates on a population of binary chromosomes, where each chromosome encodes the two decision variables x_1 and x_2 within the bounded domain [-5, 5]. The 32-bit string is split into two contiguous segments of equal length (16 bits each). Each segment is decoded to a real value using a linear min–max normalization from the integer range $[0, 2^{16} - 1]$ to the continuous interval [-5, 5].

The initial population is generated at random, producing a diverse set of candidate solutions spread across the search space. In each generation, tournament selection is used to choose parents, favouring individuals with lower objective values while still maintaining some stochasticity. Selected parents undergo one-point crossover with a specified crossover probability, producing offspring that combine genetic material from both parents. Mutation is then applied with a given mutation probability by flipping randomly chosen bits to reintroduce diversity and help escape local minima.

In every generation, the best individual is recorded, and its decoded coordinates and objective value are stored to form a trace that is later visualized on a contour plot of Himmelblau's function, illustrating the trajectory of the search towards a global minimum.

2.2 Experimental Setup

The baseline experiments are conducted using a population of 50 binary chromosomes. In every generation, tournament selection with a tournament size of 3 is used. Offspring are generated via one-point crossover with a probability of $p_c = 0.8$, and mutation is applied with a probability of $p_m = 0.01$.

To investigate the influence of the main hyperparameters on performance, a systematic parameter study is carried out. The parameters studied are mutation probability, tournament size, population size, and crossover probability. For each configuration, the algorithm is run multiple times (e.g., 10 runs) with different random seeds to account for stochastic variability.

3. RESULTS AND ANALYSIS

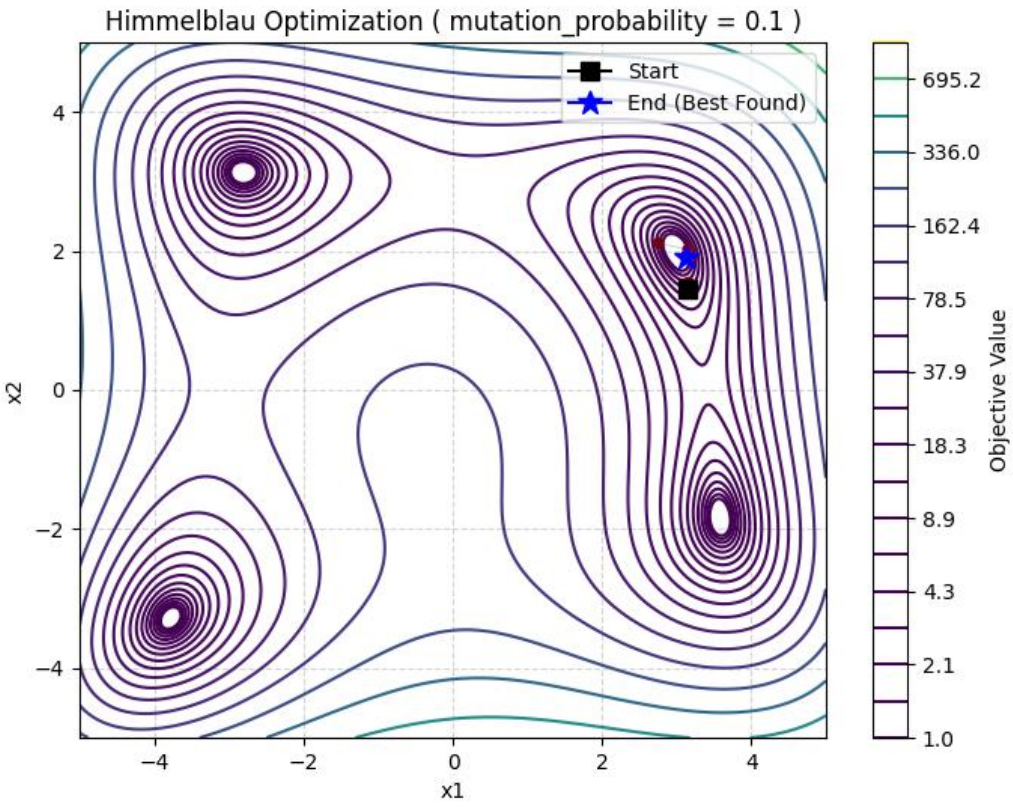
To evaluate the algorithm's performance, we analyzed the final best objective value $f(x_1, x_2)$ across 10 independent runs for each parameter setting. The mean and standard deviation for each configuration are summarized below, alongside an analysis of the search trajectory visualized on contour plots of Himmelblau's function.

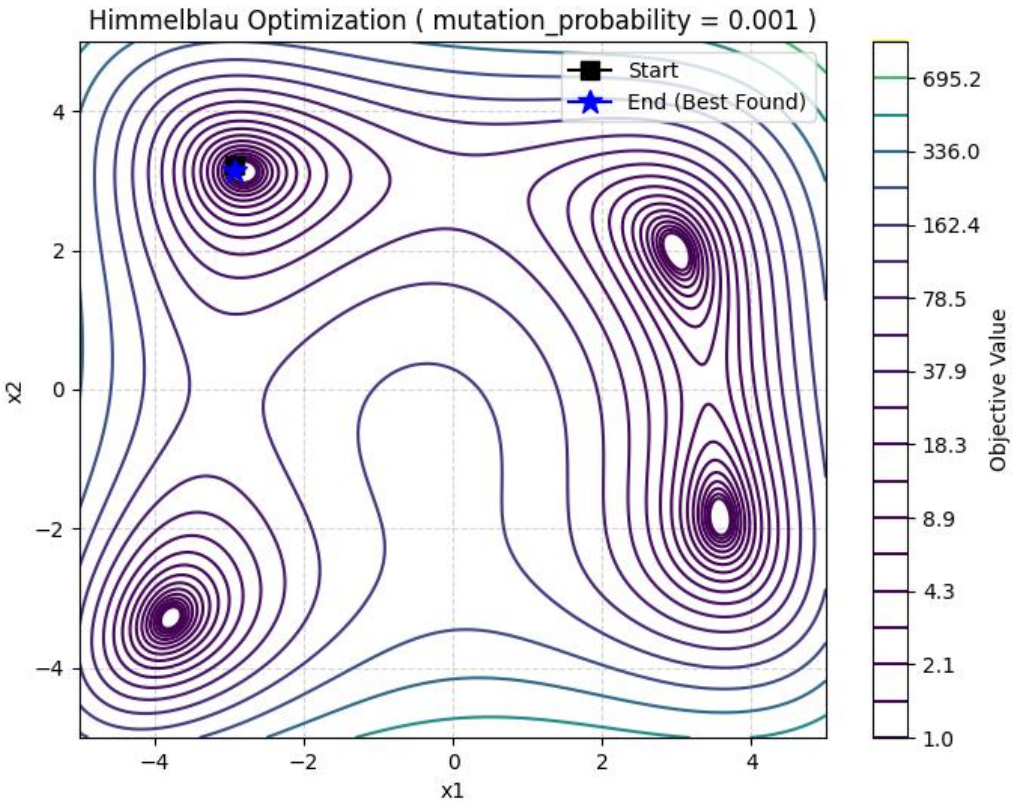
3.1 Effect of Mutation Probability (p_m)

We tested mutation probabilities of p_m in {0.001, 0.01, 0.1} while keeping other parameters at baseline (Tournament Size = 3, Population = 50, p_c = 0.8).

- **Low Mutation (p_m = 0.001):** The algorithm occasionally converged prematurely. Because very few new bits were flipped, the population struggled to escape suboptimal regions once initial diversity was lost.
  - *Mean Minimum:* 0.500374 ± 0.746019
- **Baseline Mutation (p_m = 0.01):** This provided an excellent balance between exploration and exploitation. The mutation rate was high enough to maintain diversity and prevent premature convergence, but low enough to allow the algorithm to accurately fine-tune solutions within the basin of a global minimum.
  - *Mean Minimum:* 0.018620 ± 0.018621
- **High Mutation (p_m = 0.1):** High mutation introduced too much disruption. While the algorithm explored the landscape widely, the constant bit-flipping behaved similarly to a random search, preventing the population from strictly exploiting the exact lowest points of the minima basins.
  - *Mean Minimum:* 0.329920 ± 0.444237

(Please refer to Figures below for the search trajectories corresponding to p_m = 0.001, 0.01 and 0.1 respectively).


Himmelblau Optimization (mutation_probability = 0.1)
Start
End (Best Found)
x1
x2
Objective Value
695.2
336.0
162.4
78.5
37.9
18.3
8.9
4.3
2.1
1.0


Himmelblau Optimization (mutation_probability = 0.001)
Start
End (Best Found)
x1
x2
Objective Value
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336.0
162.4
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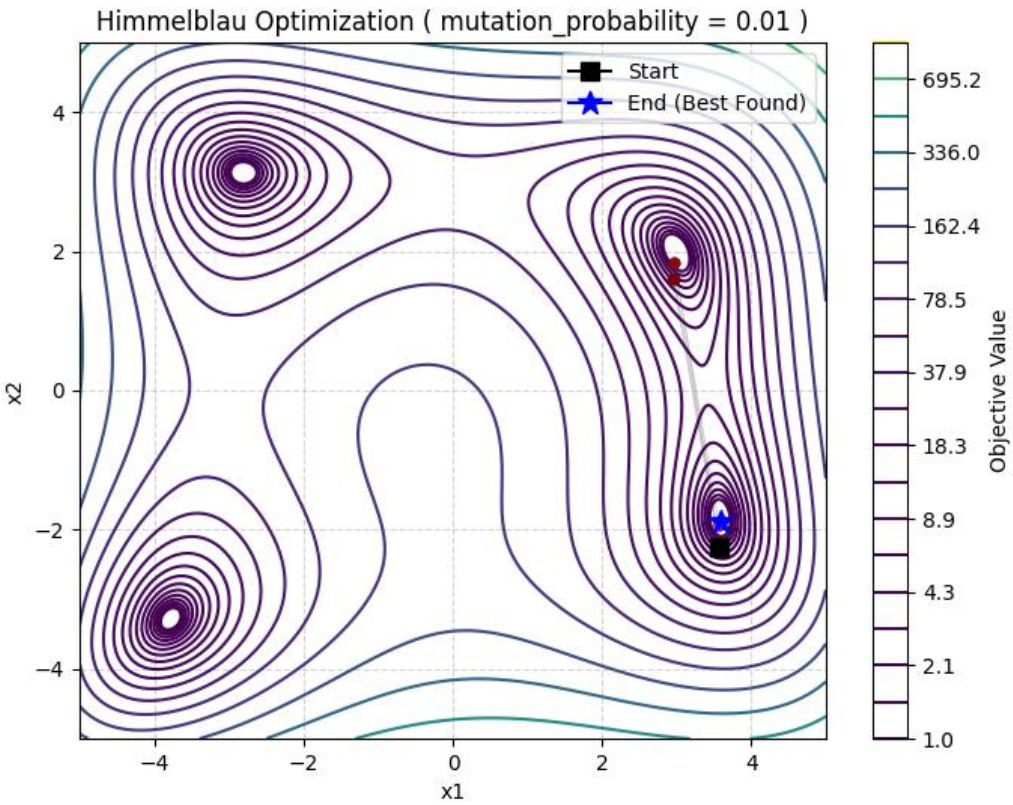
2.1
1.0

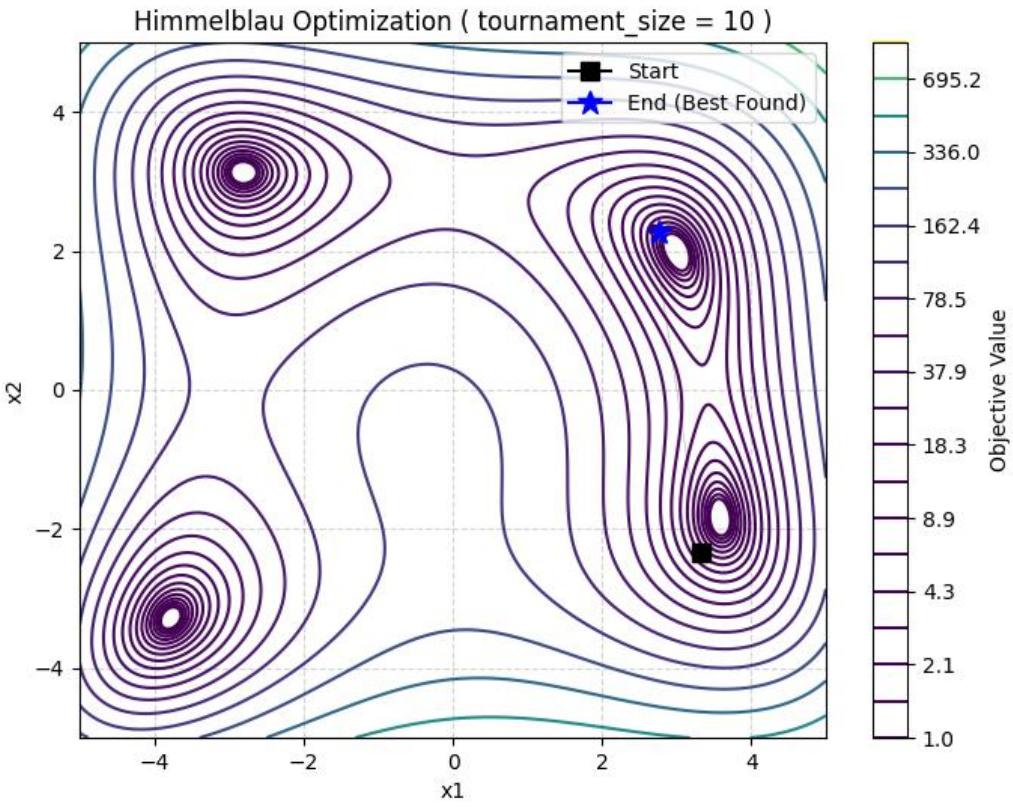


3.2 Effect of Tournament Size

We tested tournament sizes of {2, 3, 10}.

- **Tournament Size = 2:** Selection pressure was too low. The algorithm struggled to rapidly exploit good regions, leading to slower convergence and a slightly worse final approximation.
  - *Mean Minimum:* 0.258466 ± 0.367944
- **Tournament Size = 3**: This provided the optimal selection pressure for this landscape, strongly favoring fit individuals while preserving enough genetic diversity to locate the global optima.
  - *Mean Minimum:* 0.075598 ± 0.105993
- **Tournament Size = 10**: Selection pressure was overwhelmingly high. The algorithm almost immediately converged to the nearest local minimum (or a suboptimal region of a global minimum) because highly fit individuals dominated the parent pool too early, killing off diversity.
  - *Mean Minimum:* 1.112954 ± 1.679326

(Please refer to Figures below for the search trajectories corresponding to tournament sizes 2, 3, and 10 respectively).


Himmelblau Optimization (tournament_size = 10)
Start
End (Best Found)
x1
x2
Objective Value
695.2
336.0
162.4
78.5
37.9
18.3
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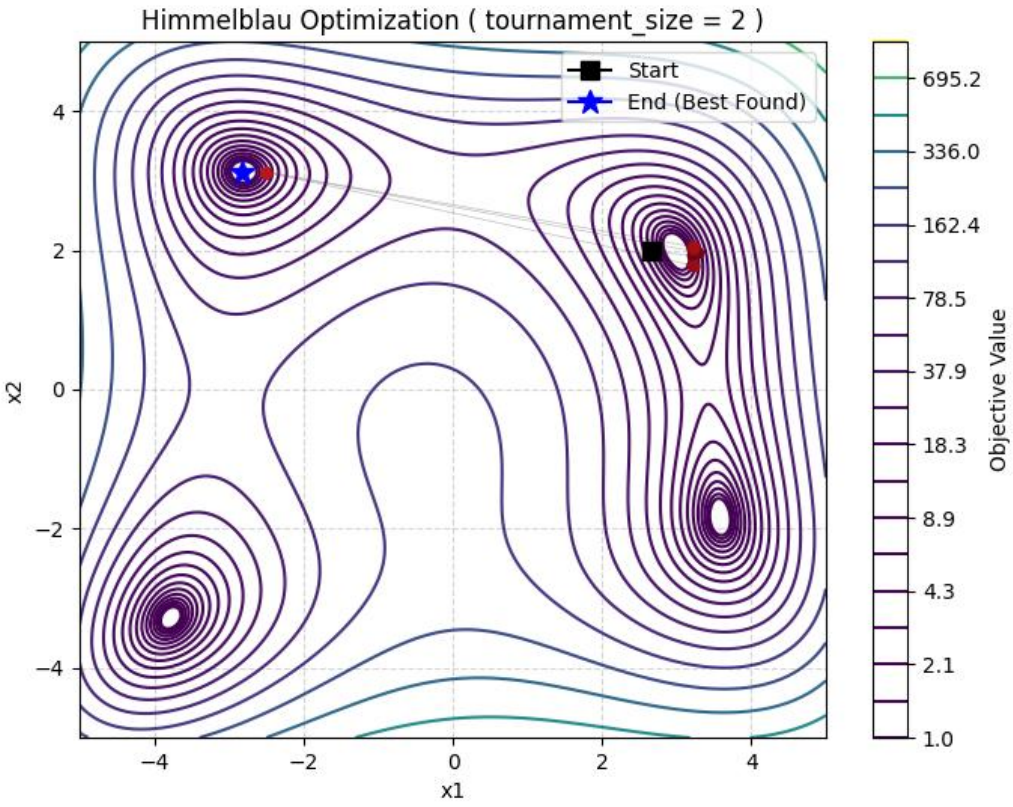
4.3
2.1
1.0


Himmelblau Optimization (tournament_size = 2)
Start
End (Best Found)
x1
x2
Objective Value
695.2
336.0
162.4
78.5
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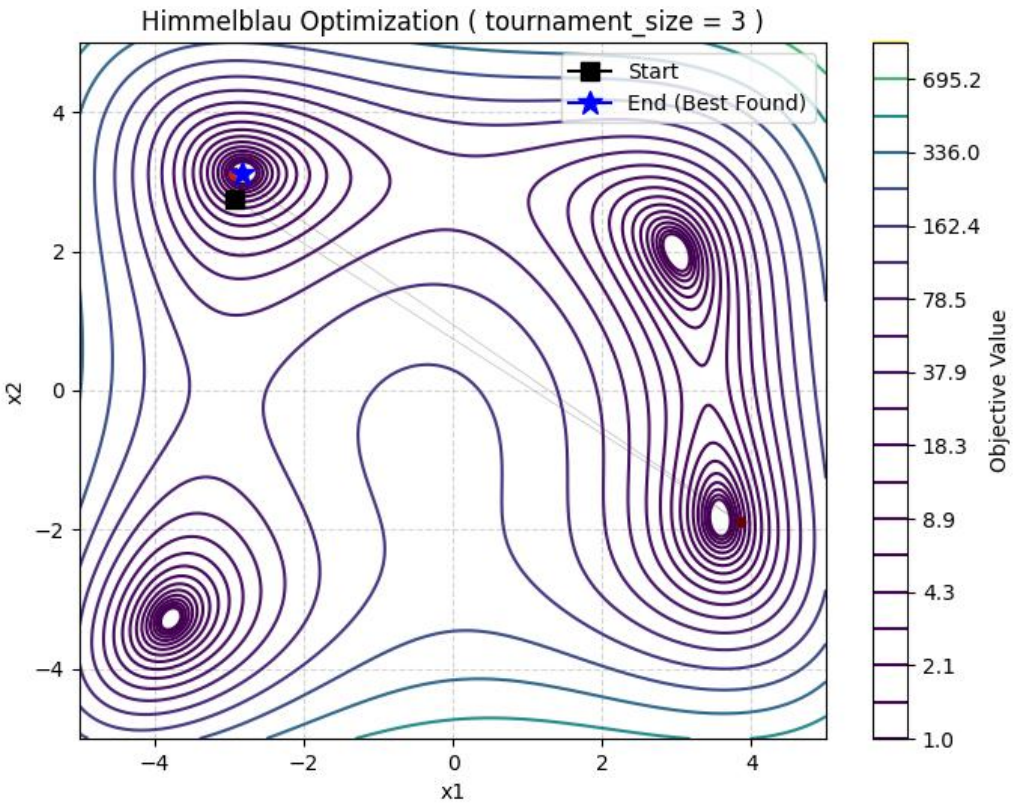
2.1
1.0

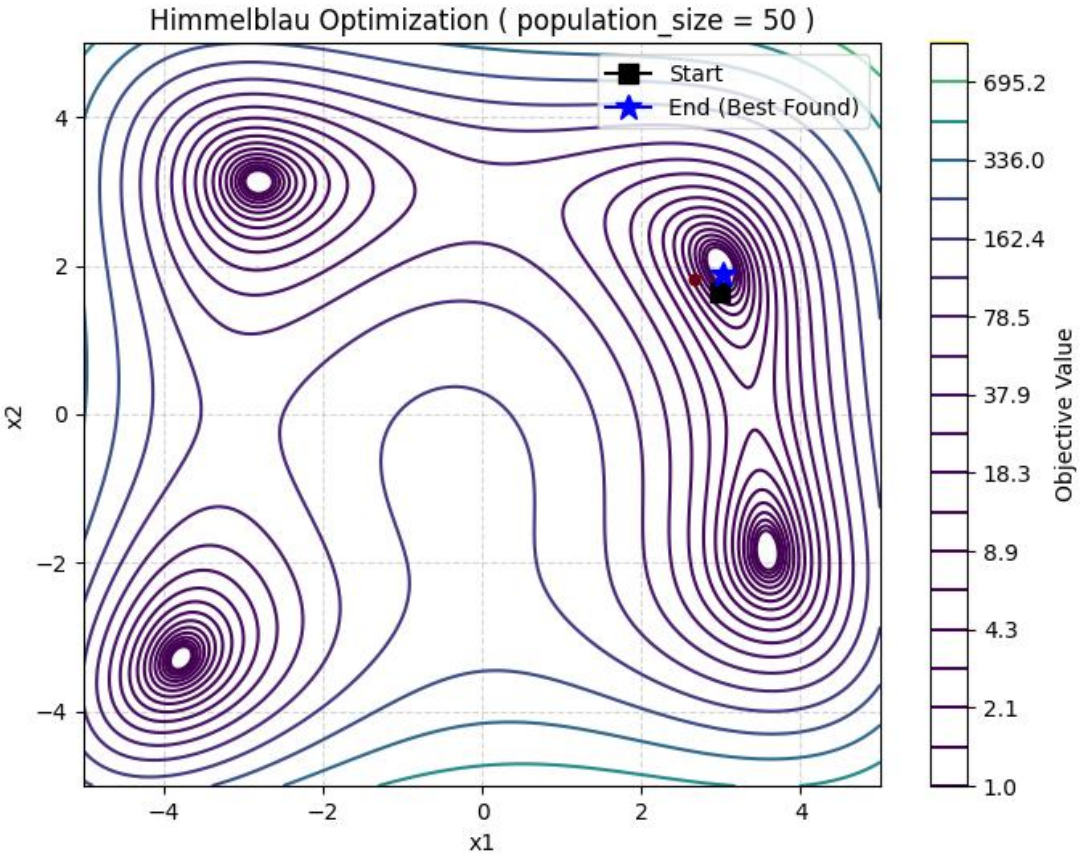


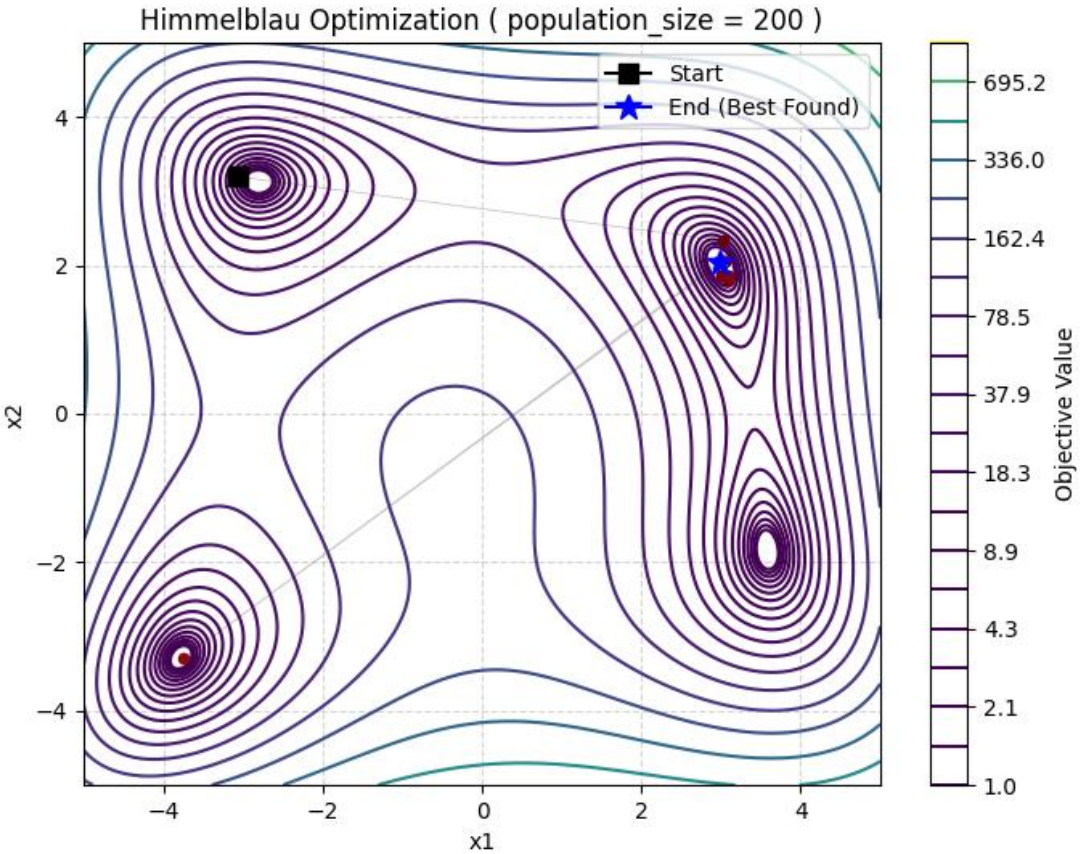
3.3 Effect of Population Size

We tested population sizes of {10, 50, 200}.

- **Small Population (10):** The algorithm failed spectacularly. A population of 10 severely lacked the initial genetic material needed to cover the multimodal landscape of Himmelblau's function, frequently becoming trapped in poor regions.
  - *Mean Minimum:* 11.807776 ± 16.452821
- **Medium Population (50):** Provided sufficient genetic diversity to reliably locate one of the global minima in almost all runs, representing a highly efficient trade-off between performance and computational cost.
  - *Mean Minimum:* 0.092340 ± 0.159281
- **Large Population (200):** The landscape was thoroughly mapped from the first generation. It achieved the most robust and accurate results, strictly exploiting the global minima, though it required significantly more objective function evaluations.
  - *Mean Minimum:* 0.057308 ± 0.150443

(Please refer to Figures below for the search trajectories corresponding to population sizes 10, 50, and 200 respectively).


Himmelblau Optimization (population_size = 50)
Start
End (Best Found)
x1
x2
Objective Value
695.2
336.0
162.4
78.5
37.9
18.3
8.9
4.3
2.1
1.0


Himmelblau Optimization (population_size = 200)
Start
End (Best Found)
x1
x2
Objective Value
695.2
336.0
162.4
78.5
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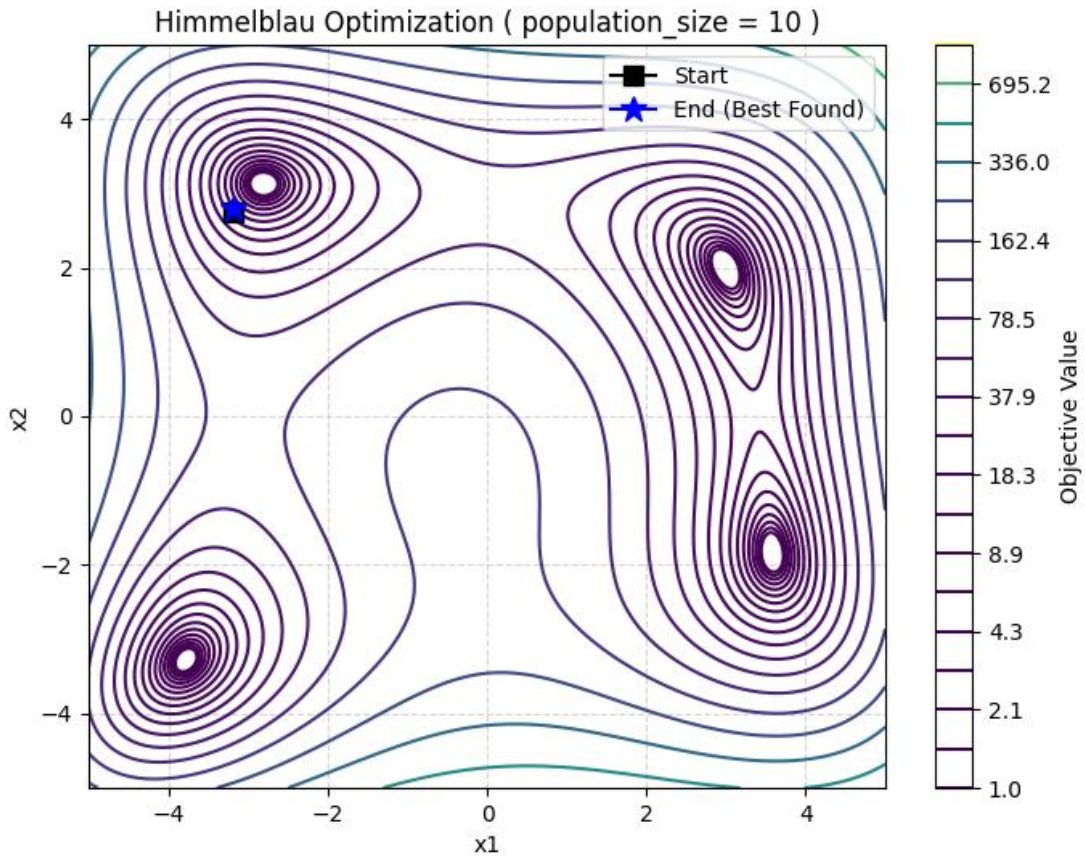
4.3
2.1
1.0

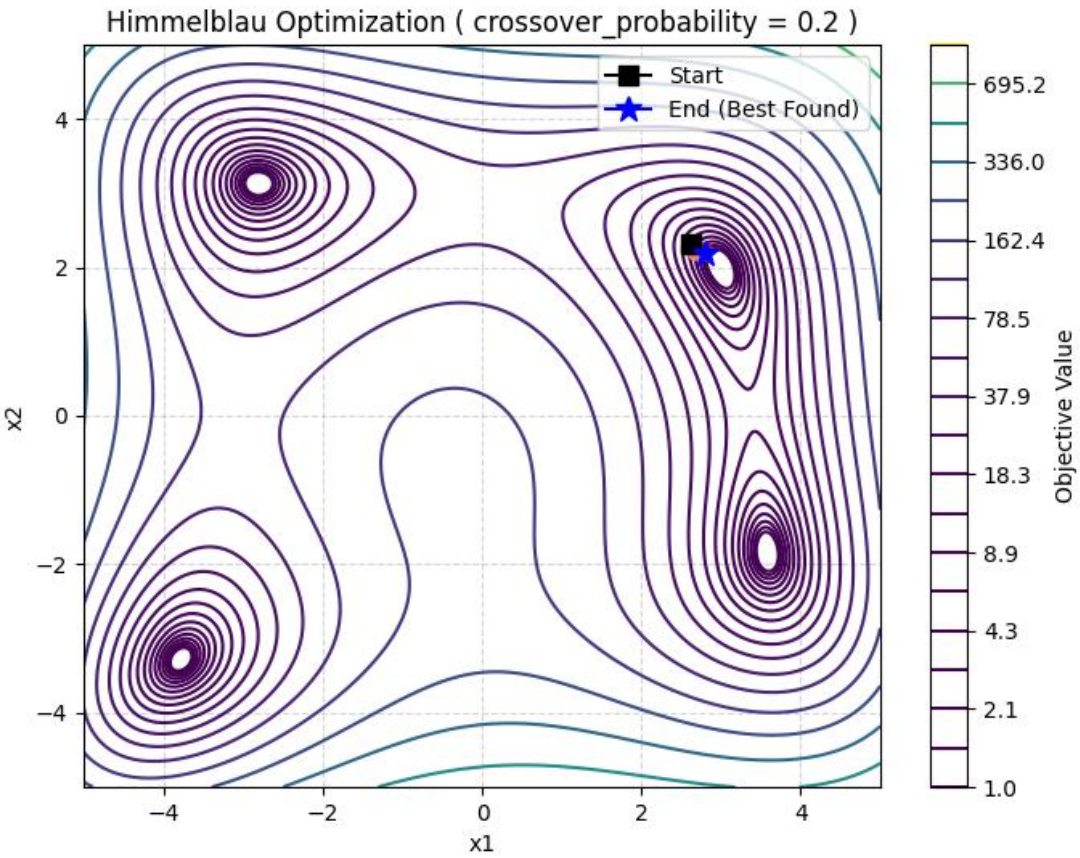


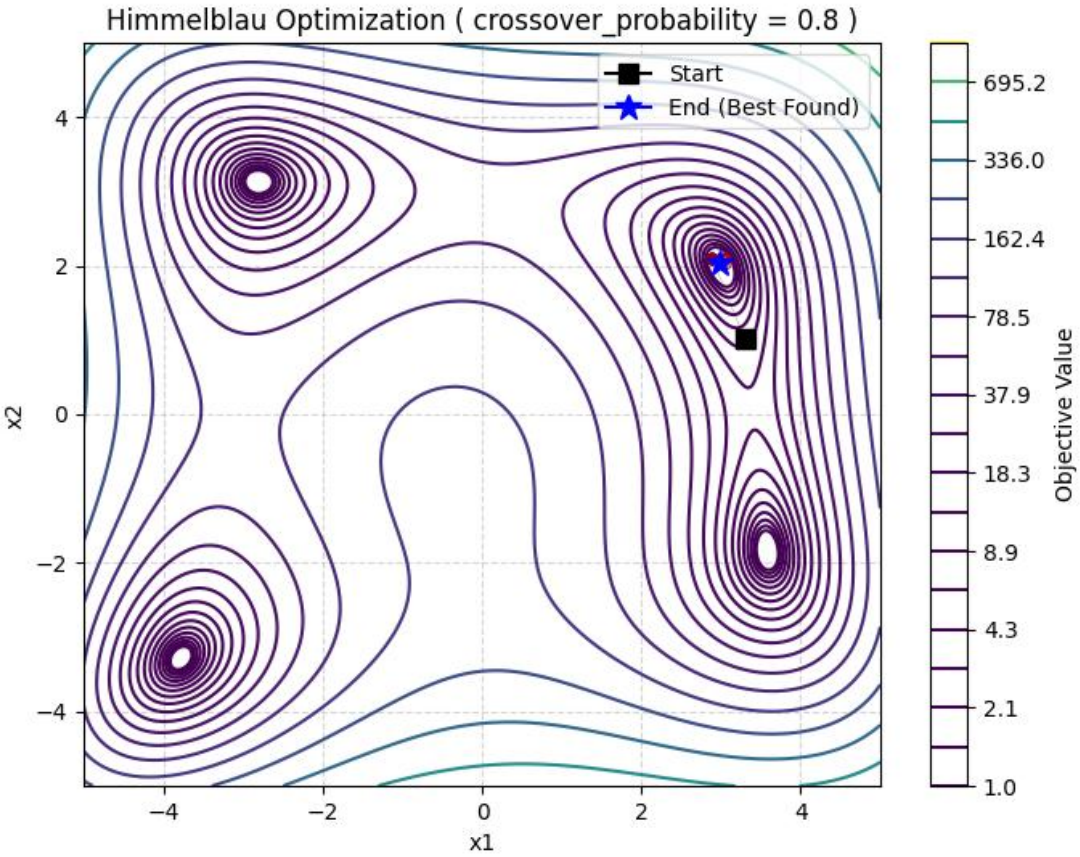
3.4 Effect of Crossover Probability (P_c)

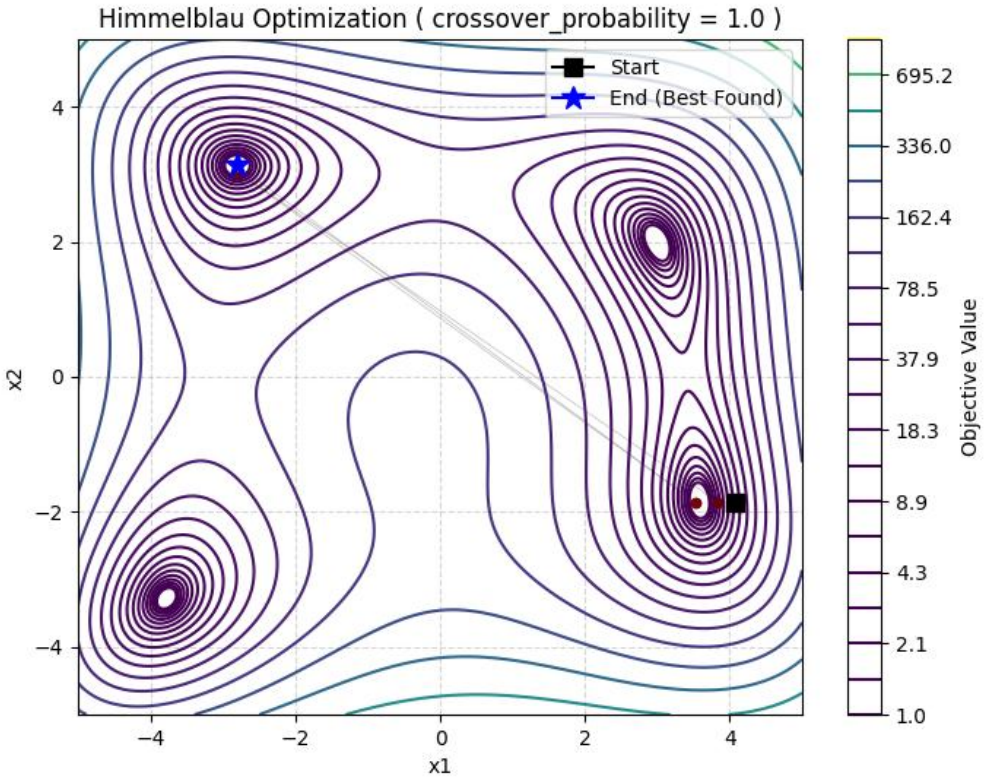
We tested crossover probabilities of p_c in {0.2, 0.8, 1.0}.

- **Low Crossover (p_c = 0.2)**: Convergence was slow and less effective. Because offspring closely resembled their parents, the algorithm relied too heavily on mutation for exploration, limiting its ability to combine advantageous traits.
  - *Mean Minimum:* 0.654878 ± 0.978657
- **Baseline Crossover (p_c = 0.8)**: Allowed for excellent mixing of genetic material, efficiently combining building blocks from fit parents to form superior offspring.
  - *Mean Minimum:* 0.198755 ± 0.360510
- **High Crossover (p_c = 1.0)**: While still effective, forcing crossover on every single parent pair occasionally disrupted highly fit individuals, leading to slightly worse mean performance than the 0.8 baseline.
  - *Mean Minimum:* 0.363861 ± 0.482086

(Please refer to Figures below for the search trajectories corresponding to p_c = 0.2, 0.8, and 1.0 respectively).


Himmelblau Optimization (crossover_probability = 0.2)
Start
End (Best Found)
x1
x2
Objective Value
695.2
336.0
162.4
78.5
37.9
18.3
8.9
4.3
2.1
1.0


Himmelblau Optimization (crossover_probability = 0.8)
Start
End (Best Found)
x1
x2
Objective Value
695.2
336.0
162.4
78.5
37.9
18.3
8.9
4.3
2.1
1.0



4. CONCLUSION

In this assignment, a binary-coded genetic algorithm was successfully implemented to minimize Himmelblau's function. The results demonstrated the algorithm's strong capability to navigate a highly non-convex, multimodal landscape. Because the objective function contains four equivalent global minima, the search naturally converged to different basins of attraction across different runs based on random initialization and stochastic genetic operators.

Our systematic parameter study highlighted the critical balance between exploration (searching new regions) and exploitation (refining known good solutions):

- An adequate population size (>50) is strictly required; dropping it to 10 resulted in catastrophic loss of diversity and a mean error of 11.8.
- Tournament size dictates selection pressure. A size of 3 was ideal, while a size of 10 induced premature convergence.
- Mutation probability acts as a fine-tuning mechanism and a safeguard against stagnation. A rate of 0.01 was optimal, yielding the best overall mean error (0.018) of the entire study, whereas higher rates disrupted convergence.
- A high crossover probability (approx. 0.8) effectively recombines advantageous genetic traits without being overly disruptive.

Ultimately, the baseline configuration (Pop=50, Tournament=3, p_m=0.01, p_c=0.8) proved to be highly robust, reliably guiding the search trajectory to one of the global optima across the vast majority of runs.